All Questions & Step-by-Step Solutions

Compiled from ChatGPT conversation

Question 1

Q: What are the eigenvalues of the matrix $\begin{pmatrix} 6 & -1 \\ 2 & 3 \end{pmatrix}$?

Solution (Step by Step):

1. The characteristic polynomial is obtained by setting

$$\det \begin{pmatrix} 6-\lambda & -1 \\ 2 & 3-\lambda \end{pmatrix} = 0.$$

2. Compute the determinant:

$$(6-\lambda)(3-\lambda) - (-1 \times 2) = (6-\lambda)(3-\lambda) + 2.$$

3. Expand:

$$18 - 6\lambda - 3\lambda + \lambda^2 + 2 = \lambda^2 - 9\lambda + 20.$$

4. Solve $\lambda^2 - 9\lambda + 20 = 0$:

$$(\lambda - 4)(\lambda - 5) = 0 \implies \lambda = 4 \text{ or } \lambda = 5.$$

Hence, the eigenvalues are $\boxed{4 \text{ and } 5}$.

Question 2

Q: Find x in the equation $3^2 - x = \frac{1}{81}$.

Solution (Step by Step):

1. Note that $3^2 = 9$. So the equation becomes

$$9 - x = \frac{1}{81}.$$

2. Isolate x:

$$-x = \frac{1}{81} - 9.$$

3. Rewrite 9 as $\frac{729}{81}$:

$$-x = \frac{1}{81} - \frac{729}{81} = \frac{1-729}{81} = \frac{-728}{81}.$$

4. Multiply both sides by -1:

$$x = \frac{728}{81}.$$

Therefore, $\boxed{x = \frac{728}{81}}$.

Question 3

Q: Evaluate the integral $\int \frac{1}{2}(x-1)^2\,dx$.

Solution (Step by Step):

1. Factor out $\frac{1}{2}$:

$$\int \frac{1}{2}(x-1)^2\,dx = \frac{1}{2}\int (x-1)^2\,dx.$$

2. Expand $(x-1)^2 = x^2 - 2x + 1$:

$$\frac{1}{2}\int (x^2 - 2x + 1)\,dx.$$

3. Integrate term by term:

$$\frac{1}{2}\left[\int x^2\,dx - \int 2x\,dx + \int 1\,dx\right].$$

$$\int x^2\,dx = \frac{x^3}{3}, \quad \int 2x\,dx = x^2, \quad \int 1\,dx = x.$$

4. Combine results:

$$\frac{1}{2}\left[\frac{x^3}{3} - x^2 + x\right] + C.$$

5. Distribute $\frac{1}{2}$:

$$\frac{x^3}{6} - \frac{x^2}{2} + \frac{x}{2} + C.$$

Hence,

$$\int \frac{1}{2}(x-1)^2\,dx = \boxed{\frac{x^3}{6} - \frac{x^2}{2} + \frac{x}{2} + C}.$$

Question 4

Q: What does $\displaystyle\int_a^b f(x)\,dx$ represent?

Solution (Conceptual):

- It represents the **net area** under the curve $f(x)$ from $x = a$ to $x = b$.
- If $f(x) \geq 0$, it's the usual area. If parts of $f(x)$ are negative, the integral subtracts that portion.
- In a physical context (e.g., if $f(x)$ is a rate of change), $\int_a^b f(x)\,dx$ gives the **accumulated quantity** over $[a, b]$.

Question 5

Q: Solve the differential equation $y'' - 10y' + 25y = 0$ where the general solution is of the form involving constants A and B.

Solution (Step by Step):

1. Write the characteristic equation:
$$r^2 - 10r + 25 = 0.$$
2. Factor (or use quadratic formula):
$$(r-5)^2 = 0 \implies r = 5 \text{ (double root)}.$$
3. For a repeated root $r = 5$, the general solution is
$$y(x) = (C_1 + C_2 x)e^{5x}.$$

Hence,
$$\boxed{y(x) = (C_1 + C_2 x)e^{5x}}.$$

Question 6

Q: In Python, consider `set = {3, 4, 2, 3, 1}`. What does `print(set)` show?

Explanation:

- A Python set *removes duplicates* and doesn't maintain a guaranteed order.
- So, $\{3, 4, 2, 3, 1\}$ becomes $\{1, 2, 3, 4\}$.
- Printing the set typically gives $\{1, 2, 3, 4\}$ (the exact order could vary, but in most Python versions, it prints sorted in this example).

Question 7

Q: Find the solution to the differential equation $y' = 3x$ with initial condition $y(1) = 3$.
Solution (Step by Step):

1. Integrate $y' = 3x$:
$$y = \int 3x\,dx = \frac{3x^2}{2} + C.$$
2. Apply initial condition $y(1) = 3$:
$$3 = \frac{3 \cdot (1)^2}{2} + C \implies C = \frac{3}{2}.$$
3. Thus:
$$\boxed{y(x) = \frac{3x^2}{2} + \frac{3}{2}}.$$

Question 8

Q: Consider the code:

```
x = 3
while x < 11:
    if x % 2 == 0:
        x = x - 1
    else:
        x = x * 2
    print(x)
```

What is its output?
Explanation (Step by Step):

1. Initially, $x = 3$. Since $3\%2 \neq 0$ (odd), x becomes $3 * 2 = 6$. Print $\boxed{6}$.
2. Now $x = 6$. $6\%2 = 0$ (even), so $x = 6 - 1 = 5$. Print $\boxed{5}$.
3. Now $x = 5$. $5\%2 \neq 0$ (odd), so $x = 5 * 2 = 10$. Print $\boxed{10}$.
4. Now $x = 10$. $10\%2 = 0$ (even), so $x = 10 - 1 = 9$. Print $\boxed{9}$.
5. Now $x = 9$. $9\%2 \neq 0$ (odd), so $x = 9 * 2 = 18$. Print $\boxed{18}$.
6. $x = 18$ is no longer < 11, loop ends.

So the output is:
$$6,\ 5,\ 10,\ 9,\ 18.$$

Question 9

Q: Express as summation: $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \frac{1}{64}$.

Solution:

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \frac{1}{64} = \sum_{k=0}^{6} \frac{1}{2^k}.$$

(There are 7 terms, from $k = 0$ to $k = 6$.)

Question 10

Q: Given the first two terms of the Taylor series expansion:

$$f(x) = f(a) + f'(a)\,(x - a) + \dots$$

what is the third term?

Solution: The Taylor series around $x = a$ is

$$f(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \frac{f'''(a)}{3!}(x - a)^3 + \cdots$$

Hence, the third term is

$$\boxed{\frac{f''(a)}{2!}(x - a)^2}.$$

Question 11

Q: What is the domain of

$$y = \sqrt{\frac{1}{x} - 1} \quad ?$$

Solution (Step by Step):

1. Inside the square root, we need $\frac{1}{x} - 1 \geq 0$.
2. So $\frac{1}{x} \geq 1$.
3. Case 1: $x > 0$. Multiply by x (positive), get $1 \geq x \implies x \leq 1$. Combine with $x > 0$ gives $0 < x \leq 1$.
4. Case 2: $x < 0$. If we multiply by x (negative), the inequality flips, giving $1 \leq x$. This contradicts $x < 0$, so no solutions there.
5. Also $x \neq 0$ because $\frac{1}{x}$ is undefined at 0.

Hence, the domain is

$$\boxed{(0,\ 1]}.$$

Question 12

Q: Let $x_0 \in (1, \infty)$ minimize $f(x) = \frac{x}{\ln(x)}$. Let $y_0 = f(x_0)$. Find $x_0 + y_0$.

Solution (Step by Step):

1. Find critical points by differentiating:

$$f(x) = \frac{x}{\ln(x)}, \quad f'(x) = \frac{(\ln(x))(1) - x\left(\frac{1}{x}\right)}{(\ln(x))^2} = \frac{\ln(x) - 1}{(\ln(x))^2}.$$

2. Set $f'(x) = 0$ implies $\ln(x) - 1 = 0 \implies \ln(x) = 1 \implies x = e$.

3. Then $f(e) = \frac{e}{\ln(e)} = \frac{e}{1} = e$. So $x_0 = e$, $y_0 = e$.

4. Therefore $x_0 + y_0 = e + e = \boxed{2e}$.

Question 13

Q: In Python:

1. $y = x$,
2. $y = x[:]$,
3. $y = \texttt{copy.deepcopy}(x)$.

Explain the difference.

Explanation:

- `y = x`: *Reference assignment.* Both x and y point to the same list in memory. Changing y changes x, and vice versa.
- `y = x[:]`: *Shallow copy.* This creates a new list object with the same top-level elements. Nested objects (e.g., sub-lists) are still shared.
- `y = copy.deepcopy(x)`: *Deep copy.* Creates a completely independent copy of x, including all nested objects.

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Questions and Solutions from Chat

ChatGPT

December 21, 2024

Introduction

This document contains a compilation of questions asked during the chat along with detailed, step-by-step solutions or explanations. Each question is numbered in the order it appeared, and the corresponding solution follows immediately.

Question 1

Question:

$$\text{Evaluate the integral } \int_0^{\frac{\pi}{2}} \cos(x)\, e^{-\sin(x)}\, dx.$$

Solution

Let us denote the integral by

$$I = \int_0^{\frac{\pi}{2}} \cos(x)\, e^{-\sin(x)}\, dx.$$

Use the substitution $u = \sin(x)$. Then $du = \cos(x)\, dx$. As x goes from 0 to $\frac{\pi}{2}$:

$$u = \sin(x) \quad \Rightarrow \quad u(0) = 0, \quad u\left(\frac{\pi}{2}\right) = 1.$$

Hence,

$$I = \int_0^1 e^{-u}\, du = \Big[-e^{-u}\Big]_0^1 = -e^{-1} + e^0 = 1 - e^{-1}.$$

Therefore,

$$\boxed{1 - \frac{1}{e}}.$$

Question 2

Question: Solve the differential equation

$$y'' + 16y = 0, \quad y(0) = 2, \quad y'(0) = 8.$$

Solution

1. Characteristic equation:
$$r^2 + 16 = 0 \quad \Longrightarrow \quad r = \pm 4i.$$
2. General solution for complex roots $\pm 4i$:
$$y(t) = C_1 \cos(4t) + C_2 \sin(4t).$$
3. Use initial conditions to find C_1, C_2.
$$y(0) = C_1 \cos(0) + C_2 \sin(0) = C_1 = 2.$$
$$y'(t) = -4C_1 \sin(4t) + 4C_2 \cos(4t).$$
$$y'(0) = -4C_1 \sin(0) + 4C_2 \cos(0) = 4C_2 = 8 \quad \Rightarrow \quad C_2 = 2.$$
Hence the solution is
$$\boxed{y(t) = 2\cos(4t) + 2\sin(4t).}$$

Question 3

Question: *"Regressing y on x means:"*

Answer (Conceptual Explanation)

Regressing y on x refers to fitting a model where y (the dependent variable) is expressed as a function of x (the independent variable). In a linear-regression context, it typically means:
$$y = \beta_0 + \beta_1 x + \epsilon,$$
where β_0 (intercept) and β_1 (slope) are parameters to be estimated, and ϵ is the error term.

Question 4

Question:
$$\text{Let } A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}. \quad \text{Find } \operatorname{rank}(A).$$

Solution

$$A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}.$$
Perform row operations:
$$R_2 \leftarrow R_2 - 2R_1 = \begin{bmatrix} 1 & 4 \\ 2 - 2 \cdot 1 & 3 - 2 \cdot 4 \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 0 & -5 \end{bmatrix}.$$
Both rows are nonzero, so the rank is 2.

$\boxed{\operatorname{rank}(A) = 2.}$

Question 5

Question: Given $P(A) = 0.4,\ P(A \mid B) = 0.2,\ P(B) = 0.6$, find $P(B \mid A)$.

Solution

By Bayes' Theorem,

$$P(B \mid A) = \frac{P(A \mid B)\, P(B)}{P(A)} = \frac{0.2 \times 0.6}{0.4} = \frac{0.12}{0.4} = 0.3.$$

Question 6

Question: Compute the dot product $[6, 5, 2] \cdot [1, 9, 4]$.

Solution

$$[6, 5, 2] \cdot [1, 9, 4] = (6 \times 1) + (5 \times 9) + (2 \times 4) = 6 + 45 + 8 = 59.$$

Question 7

Question: *"Let $x_0 \in \mathbb{R}$ be the value of x that maximizes the function $f(x) = e^{-(x^2+1)}$. Find that x_0 and the maximum value $y_0 = f(x_0)$, then compute $x_0 + y_0$."*

Solution

(a) We want to maximize $f(x) = e^{-(x^2+1)}$.

(b) Differentiate:

$$f'(x) = \frac{d}{dx}\left(e^{-(x^2+1)}\right) = e^{-(x^2+1)} \cdot \frac{d}{dx}[-(x^2+1)] = e^{-(x^2+1)} \cdot (-2x).$$

Set $f'(x) = 0 \implies -2x\, e^{-(x^2+1)} = 0$. Since $e^{-(x^2+1)} \neq 0$, we have $x = 0$.

(c) Evaluate $f(0) = e^{-(0^2+1)} = e^{-1}$.

(d) Thus $x_0 = 0,\ y_0 = e^{-1}$.

(e) Finally,

$$x_0 + y_0 = 0 + e^{-1} = e^{-1}.$$

Question 8

Question:

$$f(x) = e^{-\cos x} \sin(x). \quad \text{Find } f'\left(\tfrac{\pi}{2}\right).$$

Solution

We use the product rule: $u(x) = e^{-\cos x}$, $v(x) = \sin x$.

$$u'(x) = e^{-\cos x} \cdot \frac{d}{dx}(-\cos x) = e^{-\cos x} \sin x.$$
$$v'(x) = \cos x.$$

Hence,

$$f'(x) = u'(x)\, v(x) + u(x)\, v'(x) = \left(e^{-\cos x} \sin x\right) \sin x + \left(e^{-\cos x}\right) \cos x.$$
$$f'(x) = e^{-\cos x}\left(\sin^2 x + \cos x\right).$$

Evaluate at $x = \frac{\pi}{2}$:

$$\sin\left(\tfrac{\pi}{2}\right) = 1, \quad \cos\left(\tfrac{\pi}{2}\right) = 0.$$

So,

$$f'\left(\tfrac{\pi}{2}\right) = e^{-0}\left(1^2 + 0\right) = 1.$$

Question 9

Question: Find x such that

$$3^{2-x} = \frac{1}{81}.$$

Solution

We know $81 = 3^4$. Hence

$$\frac{1}{81} = 3^{-4}.$$

So the equation $3^{2-x} = 3^{-4}$ gives

$$2 - x = -4 \quad \Longrightarrow \quad x = 6.$$

Question 10

Question: *"For which value of x are the vectors $[\,x,\, 3]$ and $[\,12,\, -18]$ not linearly independent?"*

Solution

Two vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ in $\mathbb{R}^2$ are linearly dependent if one is a scalar multiple of the other:

$$[x,\, 3] = k\,[\,12,\, -18].$$

Comparing components:

$$x = 12k, \quad 3 = -18k.$$

From $3 = -18k$, we get $k = -\frac{1}{6}$. Then $x = 12 \cdot (-\frac{1}{6}) = -2$. Hence the vectors are linearly dependent (i.e., not independent) if $x = -2$.

Question 11

Question: *"Given the first two terms in the Taylor series, what is the third term?*

$$f(x) = f(a) + f'(a)(x-a) + \dots$$

"

Solution (General Taylor Series Form)

A Taylor expansion of $f(x)$ around a is:

$$f(x) = f(a) \;+\; f'(a)\,(x-a) \;+\; \frac{f''(a)}{2!}(x-a)^2 \;+\; \dots$$

Hence the *third term* (the one involving $(x-a)^2$) is

$$\frac{f''(a)}{2!}\,(x-a)^2.$$

Question 12

Question: *"What is the probability of getting no prime numbers if 2 dice are thrown?"*

Solution

- The prime faces on a standard die are $\{2, 3, 5\}$.
- The non-prime faces are $\{1, 4, 6\}$.
- Each die has 6 faces, so there are $6 \times 6 = 36$ total outcomes.
- We want both dice to show a non-prime face. The non-prime faces are 1,4,6 (3 options out of 6). Number of favorable outcomes $= 3 \times 3 = 9$.
- Probability:

$$\frac{9}{36} = \frac{1}{4}.$$

Question 13

Question: Given $y' = 6x^2 - 2$ and the point $(2, 22)$, find $y(x)$.

Solution

$$y' = 6x^2 - 2 \implies y = \int (6x^2 - 2)\,dx = \int 6x^2\,dx - \int 2\,dx = 2x^3 - 2x + C.$$

Use $(x, y) = (2, 22)$ to solve for C:

$$22 = 2(2)^3 - 2(2) + C = 2 \cdot 8 - 4 + C = 16 - 4 + C = 12 + C \implies C = 10.$$

Hence

$$\boxed{y = 2x^3 - 2x + 10.}$$

Question 14

Question: *"What does the following expression give?* $\int_a^b f(x)\,dx$*"*

Solution (Interpretation)

$\int_a^b f(x)\,dx$ is the **definite integral** of f from a to b. It represents:

- The signed area under the curve $y = f(x)$ from $x = a$ to $x = b$.
- More generally, the net accumulation of the quantity described by f over $[a, b]$.

Question 15

Question: *"Simplify the infinite series* $1 - x + x^2 - x^3 + \cdots$*."*

Solution

This is the sum of an infinite geometric series with first term 1 and common ratio $-x$. For $|x| < 1$, its sum is given by

$$\frac{1}{1-(-x)} = \frac{1}{1+x}.$$

Hence

$$1 - x + x^2 - x^3 + \cdots = \frac{1}{1+x}, \quad \text{for } |x| < 1.$$

Question 16

Question: $\displaystyle\sum_{i=1}^{\infty} \frac{1}{2^4}$.

Solution

$$\frac{1}{2^4} = \frac{1}{16}.$$

The series becomes

$$\sum_{i=1}^{\infty} \frac{1}{16}.$$

Each term is a constant $\frac{1}{16}$. Such a series diverges, since adding $\frac{1}{16}$ infinitely many times grows without bound. Formally,

$$\sum_{i=1}^{\infty} (\text{constant}) \;\to\; \infty.$$

Thus it does *not* converge to a finite sum.

Question 17

Question: *"Simplify* $\sin^2 x + \cos^2 x$*."*

Solution

This is a fundamental Pythagorean identity:

$$\sin^2 x + \cos^2 x = 1.$$

Question 18

Question: *"Given* $f(x, y) = x + y$ *subject to* $x^2 + y = 1$*. At which point does the maximum of* f *occur?"*

Solution (Method of Lagrange Multipliers)

$$f(x, y) = x + y, \quad g(x, y) = x^2 + y - 1 = 0.$$

The Lagrangian:

$$\mathcal{L}(x, y, \lambda) = x + y + \lambda(x^2 + y - 1).$$

Take partial derivatives and set equal to zero:

$$\frac{\partial \mathcal{L}}{\partial x} = 1 + \lambda \cdot 2x = 0, \quad \frac{\partial \mathcal{L}}{\partial y} = 1 + \lambda = 0, \quad \frac{\partial \mathcal{L}}{\partial \lambda} = x^2 + y - 1 = 0.$$

From $\frac{\partial \mathcal{L}}{\partial y} = 1 + \lambda = 0$, we get $\lambda = -1$. Then $\frac{\partial \mathcal{L}}{\partial x} = 1 + (-1) \cdot 2x = 1 - 2x = 0 \implies x = \frac{1}{2}$. Constraint $x^2 + y = 1 \implies \left(\frac{1}{2}\right)^2 + y = 1 \implies \frac{1}{4} + y = 1 \implies y = \frac{3}{4}$. Hence the maximum occurs at $\left(\frac{1}{2}, \frac{3}{4}\right)$. The maximum value is

$$f\left(\tfrac{1}{2}, \tfrac{3}{4}\right) = \tfrac{1}{2} + \tfrac{3}{4} = \tfrac{5}{4}.$$

Question 19

Question: *"What is the output of the Python code:* $(1 + 2 * *2) * *(1 + 1) * 2$*?"*

Solution

$$2 * *2 = 4 \implies 1 + 4 = 5.$$

Then $(1 + 2 * *2)^{(1+1)} = (5)^2 = 25$. Finally multiply by 2:

$$25 \times 2 = 50.$$

Hence the output is $\boxed{50}$.

Question 20

Question: *"What are the eigenvalues of the matrix* $\begin{bmatrix} 6 & -1 \\ 2 & 3 \end{bmatrix}$*?"*

Solution

Solve $\det(A - \lambda I) = 0$. For

$$A - \lambda I = \begin{bmatrix} 6-\lambda & -1 \\ 2 & 3-\lambda \end{bmatrix}.$$

$$\det(A - \lambda I) = (6-\lambda)(3-\lambda) - (-1)(2) = (6-\lambda)(3-\lambda) + 2.$$

Expand:

$$(6-\lambda)(3-\lambda) = 18 - 6\lambda - 3\lambda + \lambda^2 = \lambda^2 - 9\lambda + 18,$$

so

$$\det(A - \lambda I) = \lambda^2 - 9\lambda + 18 + 2 = \lambda^2 - 9\lambda + 20.$$

We solve

$$\lambda^2 - 9\lambda + 20 = 0.$$

Factor:

$$(\lambda - 4)(\lambda - 5) = 0 \implies \lambda = 4 \text{ or } \lambda = 5.$$

Question 21

Question: *"How many combinations are there of choosing 2 items from 12?"*

Solution

Use $\binom{n}{r} = \frac{n!}{r!(n-r)!}$ with $n = 12$ and $r = 2$:

$$\binom{12}{2} = \frac{12!}{2!(12-2)!} = \frac{12!}{2!\,10!}.$$

Since $12! = 12 \times 11 \times 10!$,

$$\binom{12}{2} = \frac{12 \times 11}{2 \times 1} = 66.$$

Question 22

Question:

$$\text{Evaluate } \int \ln(2x)\,dx.$$

Solution (Integration by Parts)

Let $u = \ln(2x)$ and $dv = dx$. Then $du = \frac{d}{dx}[\ln(2x)]\,dx = \frac{1}{2x} \cdot 2\,dx = \frac{1}{x}\,dx$, and $v = \int dx = x$.

Integration by parts formula:

$$\int u\,dv = uv - \int v\,du.$$

Thus

$$\int \ln(2x)\,dx = x\ln(2x) \;-\; \int x \cdot \frac{1}{x}\,dx = x\ln(2x) \;-\; \int 1\,dx = x\ln(2x) - x + C.$$

Hence

$$\boxed{\int \ln(2x)\,dx = x\ln(2x) - x + C.}$$

Final Note

This document includes the step-by-step solutions to each of the questions posed in the chat. To generate a PDF:

(a) Copy this entire LaTeX code into a file named `chat_qa.tex`.

(b) Use a LaTeX compiler (such as `pdflatex chat_qa.tex`) locally, or use an online platform like Overleaf.

(c) A PDF document containing all questions and solutions will be generated.

Chat Transcript: Questions and Step-by-Step Solutions

Generated by ChatGPT

December 21, 2024

Questions and Solutions

1. What is the equivalent statement for $\sin(2\theta)$**?**
Solution: Using the double-angle formula:

$$\sin(2\theta) = 2 \cdot \sin(\theta) \cdot \cos(\theta)$$

2. What is the median of the set (1, 2, 5, 6, 6)?
Solution: The median is the middle value in the sorted set:

$$(1, 2, 5, 6, 6)$$

The middle value is 5. Hence, the median is 5.

3. Python has all the following characteristics except: object-oriented, interpreted, interactive, compiled.
Solution: Python is object-oriented, interpreted, and interactive, but it is not compiled. Thus, the correct answer is **compiled**.

4. How many combinations are there of choosing 7 items from 9?
Solution: Using the formula for combinations:

$$C(n, k) = \frac{n!}{k! \cdot (n-k)!}$$

For $n = 9$ and $k = 7$:

$$C(9, 7) = \frac{9!}{7! \cdot 2!} = 36$$

5. What is the probability of getting the sum of two 6-sided dice as a prime number?
Solution: Prime numbers possible: $2, 3, 5, 7, 11$.
Number of favorable outcomes: 15.

Total outcomes: 36.
Probability:

$$P = \frac{15}{36} = \frac{5}{12}$$

6. What is the output of the code?

```
A = {1, 2, 3}
B = {1, 3, 4, 5}
C = {1, 2, 5, 7}
print(A & B | C)
```

Solution: Step-by-step:

$$1.\, A \cap B = \{1, 3\}$$
$$2.\, (A \cap B) \cup C = \{1, 2, 3, 5, 7\}$$

Output: $\{1, 2, 3, 5, 7\}$.

7. Evaluate the partial derivative $\frac{\partial f}{\partial x}$ for $f(x, y) = 2x^2 - 3xy + 3y^2$.
Solution:

$$f(x, y) = 2x^2 - 3xy + 3y^2$$
$$\frac{\partial f}{\partial x} = 4x - 3y$$

8. If $P(A) = 0.4$, $P(A|B) = 0.2$, and $P(B) = 0.6$, find $P(B|A)$.
Solution: Using Bayes' Theorem:

$$P(B|A) = \frac{P(A|B) \cdot P(B)}{P(A)}$$
$$P(B|A) = \frac{0.2 \cdot 0.6}{0.4} = 0.3$$

9. How many permutations are there of the letters in 'ABCD'?
Solution: For 4 distinct letters, the number of permutations:

$$P = 4! = 24$$

10. Evaluate: $y'' + 16y = 0$, where A and B are arbitrary constants.
Solution: The characteristic equation:

$$r^2 + 16 = 0$$
$$r = \pm 4i$$

General solution:

$$y(t) = A\cos(4t) + B\sin(4t)$$

Questions and Solutions

December 21, 2024

Introduction

This document contains all the questions and solutions discussed, presented with step-by-step explanations.

Questions and Solutions

Q1

For the resulting distribution to be approximately normal, the Central Limit Theorem requires the distribution of the sum of a larger number of:

- I. dependent distributions
- II. identical distributions
- III. underlying binomial variables

Solution: The correct answer is **II. identical distributions**.

Q2

If $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$, what is the rank of matrix A?
Solution: The rank of matrix A is 2.

Q3

Which of the following is an independent event?

- Pick a card. Put it back. Pick another card.
- Picking a king given the first card was a queen.
- Pick a card. Then pick another card.
- None of these.

Solution: The independent event is **Pick a card. Put it back. Pick another card.**

Q4

If a student got 75 points on an exam where the average score was 60 and the standard deviation was 10, what is the z-score of their result?
Solution:

$$\begin{aligned}\text{z-score} &= \frac{x-\mu}{\sigma} \\ &= \frac{75-60}{10} \\ &= 1.5\end{aligned}$$

The z-score is **1.5**.

Q5

What is the output for this line of Python code: $12 + 2 \times 24 \div 3 \times 4 + 4$?
Solution: The output is **80**.

Q6

Which of the following functions is continuous at $x = 0$?

- $y = \begin{cases} \sin(x) & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$
- $y = \begin{cases} \sin(x) & \text{if } x \leq 0 \\ x & \text{if } x > 0 \end{cases}$
- $y = \begin{cases} x+1 & \text{if } x \leq 0 \\ -x-1 & \text{if } x > 0 \end{cases}$
- None of these.

Solution: The function that is continuous at $x = 0$ is:

$$y = \begin{cases} \sin(x) & \text{if } x \leq 0 \\ x & \text{if } x > 0 \end{cases}$$

Q7

What is the determinant of $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 4 & 5 \\ 3 & 2 & 1 \end{bmatrix}$?
Solution: The determinant of the matrix is **-12**.

Q8

Which of the following vectors is linearly independent with $[1, 3, 0]$ and $[2, 0, 2]$?
Solution: The vector $[0, 0, 1]$ is linearly independent with the given vectors.

Q9

Find x: $x^3 = -8$.
Solution:

$$\begin{aligned} x &= \sqrt[3]{-8} \\ x &= -2 \end{aligned}$$

The solution is $x = -2$.

Q10

If $P(A) = 0.4$, $P(A|B) = 0.2$, and $P(B) = 0.6$, then find $P(B|A)$.
Solution:

$$\begin{aligned} P(B|A) &= \frac{P(A|B) \cdot P(B)}{P(A)} \\ &= \frac{0.2 \cdot 0.6}{0.4} \\ &= 0.3 \end{aligned}$$

The answer is $P(B|A) = 0.3$.

Q11

What is the probability of picking 2 kings when selecting 2 cards from a 52 card deck?
Solution:

$$\begin{aligned} P(\text{Two Kings}) &= \frac{4}{52} \cdot \frac{3}{51} \\ &= \frac{1}{221} \end{aligned}$$

The probability is approximately 0.0045.

Q12

Evaluate:

$$\lim_{x \to 4} \frac{x^2 - 16}{x - 4}$$

Solution:

$$\begin{aligned}\frac{x^2-16}{x-4} &= \frac{(x-4)(x+4)}{x-4} \\ &= x+4 \quad (\text{for } x \neq 4) \\ \lim_{x\to 4} f(x) &= 4+4=8\end{aligned}$$

The limit is **8**.

Q13

If $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 4 & 5 \\ 3 & 2 & 1 \end{bmatrix}$, what is the rank of matrix A?

Solution: The rank of matrix A is **2**.